

Time study examining the effect of range, cage-free, and cage environments on man-hours committed to bird care in 3 brown egg layer strains

Kenneth E. Anderson¹

Department of Poultry Science, North Carolina State University, Raleigh 27695-7608

Primary Audience: Egg Producers, Complex Managers, Flock Supervisors, Researchers

SUMMARY

Growing consumer awareness concerning layer hen welfare has caused the egg industry to consider alternative environments for egg-laying hens. However, the time commitment for alternative care has not been researched in detail since the 1950s. In 2 experiments, we evaluated the man-hour commitment associated with 3 different environments, including range (R), cage-free (CF), and cage systems (C). Concurrently within the R and CF systems, 3 strains of birds, Hy-Line Silver Brown, Hy-Line Brown, and Barred Plymouth Rock, were examined to see if strain differences were associated with man-hours associated for care. All birds were raised within the environment they were going to be laying in at 17 wk of age, the time data collection started. The system study was a 3-factor randomized design that commenced at 17 wk, when the laying phase commenced with 8 R replicates, 24 CF replicates, and 4 C replicates. The strain study consisted of R and CF systems and 3 strains of Hy-Line Brown, Hy-Line Silver Brown, and Barred Plymouth Rock. All husbandry was provided as appropriate to the production system and in accordance with the approved animal care procedures. Time was recorded for all of the procedures done within the replicates (i.e., egg collection, feeding, and so on). The data were transformed first by man-hours per bird housed and second by man-hours per bird surviving. All time data were analyzed using PROC GLM in SAS. Man-hours per hen decreased from 17 to 37 wk in all production systems. The production environment C required only 0.334 h/hen housed, which is lower as compared with either the CF at 0.486 h/hen housed or R at 1.268 h/hen housed. Strain alone did not influence man-hours; however, the strain with the poorest livability had the greatest man-hour requirement for hens surviving. This study substantiates previous findings that moving from intensive to extensive production systems will increase man-hours per hen by 3 or 4 times.

Key words: brown egg layer, heritage layer, range, cage-free, cage, labor

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DESCRIPTION OF PROBLEM

Alternative production systems of cage-free or range egg production are increasing in number due to outside influences [1]. The outside

constituents forcing this growth are creating great change, and with these changes come uncertainty in relation to the future for intensive egg production. Concerns still exist regarding the effect of the cage environment on laying hen

¹Corresponding author: ken_anderson@ncsu.edu

well-being, but along with this are concerns related to the cage-free and range systems as they relate to the time commitment to actually care for the birds. Labor inputs for egg production have always been of concern, and means of minimizing labor has been discussed in early works [2–4]. Those authors indicated that labor varied across season and unit size, but indicated that man-hours per bird ranged from about 2.2 to 2.6 h/hen. To examine labor with modern commercial layers, a preliminary study by Anderson [5] tracked the man-hours to care for free-range hens and hens in cage production facilities. In this preliminary study it was found that within the North Carolina Layer Performance and Management Test, through an 85-wk production cycle, hens on the range required 2.663 man-hours/hen, whereas the caged hens only required 0.173 man-hours/hen. The man-hours for the range hens were consistent with previous works [2–4]. Whereas large producers are dealing with cage versus cage-free systems, small producers, with flocks ranging in size from 100 to 3,000 hens using commercial and heritage strains of chickens, are increasing egg production in both cage-free and range settings. Still, one of the current issues is that our knowledge base is small as to how these alternative production methods influence man-hours. Current information was collected with specific pure breeds and modern lines of poultry selected for very high rates of egg production [1]. Despite the degree of environmental protection and economic feasibility offered by traditional poultry houses, the practice of housing laying hens in cages has the public perception of providing poor welfare for laying hens with little or no perspective of the effect on hens in cage-free or range settings. As a result of increasing public interest in laying hen welfare, alternative management systems, including cage-free and free-range environments, have been developed. The objective of this study was to examine the man-hour requirements between 3 housing systems using 2 commercial strains and 1 heritage strain throughout a single production cycle.

MATERIALS AND METHODS

To examine differences in man-hours for production systems and breed influences on labor,

2 studies were conducted concurrently looking at the production environments of caged, cage-free, and range environments, with the second study examining the influence of breed on man-hours in the cage-free and free-range systems. The birds were hatched and grown at the Piedmont Research Station [6] in accordance with the methods used for the North Carolina Layer Performance and Management Test [7] and the North Carolina State University Institutional Animal Care and Use Committee approval including that for range production. The hens were reared in a manner that facilitated their learning the function of the many aspects of the laying environment for cage-free and free-range systems. Within the rearing environment it is imperative that pullets learn to properly use the facilities in the production environments, such as range, roosts, and nests for egg production, and foraging behaviors to ensure productivity of the flock. All of the brown layers were reared in the cage-free and free-range systems depending on the production for which they were intended (i.e., free range or cage free). The chicks were hatched concurrently then identified as to the production environment, range and cage free, they were destined for. The birds were brooded in comingled floor pens on litter at 689 cm²/bird. The range pullets were moved to the range after brooding to complete their rearing. The birds were maintained on the same rearing dietary regimen, vaccination, and supplemental lighting program throughout the rearing program; however, at 12 wk, the range birds went on natural light supplemented with controlled day lengths. The following procedures were conducted in accordance with the research protocol. The typical procedures included in the time study were production data recorded daily: eggs, mortality, feed distribution by weight as needed with weigh backs taken at the close of each 28-d period, egg quality data collected and recorded once every period, manure removal as needed, maintenance, and rotating paddocks

Experiment 1

The time study examining production environments used range, cage-free, and cage production, described in Table 1, as follows using commercial brown egg strains. The strains were

Table 1. Replicate and hen numbers associated with experiments 1 and 2

Item	Time study replicate	Cage sets/ replicate	Hens/ replicate	Total hens housed
Experiment 1				
House 4, cage	2	36	738	1,476
House 5, cage	2	36	756	1,476
House 2, cage free	20	1	216	4,320
Hut 1 and 2, free range	8	1	75	600
Experiment 2				
House 2, cage free	8	1	216	1,728
Hut 1 and 2, free range	6	1	75	450

the same for the North Carolina Layer Performance and Management Test [8].

Range Layout Description. The range layout consisted of 2 range huts, each consisting of 4 pens with associated paddocks. Hens were provided a minimum of 929 cm²/hen in the range hut in pens that were all slats. Hens were provided 13 cm of roosting space per hen and 1 nest per 8 hens. The range hut had a timer and light powered via battery and solar panels and a supplemental propane heater for winter conditions to maintain an interior temperature above 7.2°C (45°F), which is the lower level of the chickens’ effective thermal neutral zone where body temperature will be maintained via a feed intake increase. The hens had free access to the outdoors throughout the day and night, but appeared to return to the range hut during the dark for roosting and protection. Husbandry, lighting, and supplemental feed were allocated on the same basis as flock mates in cages to minimize the variables between flock mates as much as possible. Range density was based upon a 500 hen/acre static equivalency of 8.04 m²/hen (86 ft²/hen). The range pens were 21.3 × 21.3 m (70 × 70 ft) and were enclosed by a 1.8-m (6-ft) fence with the lower chain link section being 1.2 m (4 ft) high. To facilitate range forage replenishment, each of the paddocks were divided in half with a diagonal fence providing 4.04 m²/hen (43 ft²/hen) and rotated every 4 wk. One week before rotation the paddocks were mowed to an approximate height of 15 cm (6 in). Hen movement was controlled by an access a gate that allowed access to one-half of the paddock at any point in time. The veranda area was a 3.04- × 4.6-m (10- × 15-ft) shaded area of bare dirt. Each range hut had 8 nipple drinkers inside each pen and 8 nipple drinkers outside (turned off in winter to prevent

freezing). Tube feeders were available in each pen: 1 inside and a covered feeder outside providing 5.1 cm of feeder space per hen.

Cage-Free Layout. The cage-free layout consisted of 24 cage-free pens that were a slat-litter floor combination. The slat-litter facility pens were 3.7 × 5.5 m (12 × 18 ft) with two-thirds slats and one-third litter for a total of 20.4 m² (216 ft²). The hens were housed at approximately 929 cm² (144 in²) per hen for a total hen population of 216 hens/pen. Roosts were provided at 13 cm/hen and 1 nest per 8 hens, which provides adequate use of nests by the hens. Heat for winter conditions was provided by hen body heat, which maintained an interior temperature above 12.7°C (55°F), which is the lower level of the chickens’ thermal neutral zone where body temperature was maintained without a feed intake increase.

Cage Layout. The cage layout consisted of 2 cage production houses, with each house containing 2 rows of commercial layer cages used in this study. In both houses, each side of a bank was designated as a row and each row was divided into 36 eight-foot cage groups with 9 groups on each of 4 levels. The replicates were equipped with feed hoppers to supply and monitor feed consumption for each individual replicate and the feed was distributed by an automatic feeding system. House 4 was a high rise, environmentally controlled laying facility with 2 rows of a quad-deck (4-tier) cage system. The replicate blocks contained cages that were either 61 or 81 cm wide. Two replicate banks of cages were used with a total of 1,476 hens. House 5 was a standard height, totally enclosed, force-ventilated laying house with a scraper pit manure-handling system. It had 2 rows of a quad-deck (4-tier) cage system. The replicate

blocks contained cages that were either 61 or 81 cm wide. Two replicate banks of cages were used with a total of 1,476 hens. Heat for winter conditions was provided by hen body heat, which maintained an interior temperature above 12.7°C (55°F), which is the lower level of the chickens' thermal neutral zone where body temperature was maintained without a feed intake increase.

Experiment 2

In this experiment, the evaluated man-hour commitment associated with 2 commercial strains of brown egg layers [Hy-Line Silver Brown (**HBS**) and Hy-Line Brown (**HB**)] and a heritage strain [Barred Plymouth Rock (**BPR**)] were examined. Man-hour commitments were collected such that strain and production system environments differences were elucidated. Environments included the range and cage-free systems. Within the range system, 6 replicates of 75 hens (2 reps for each strain/breed) were used; the cage-free system consisted of 8 replicates of 216 hens (3 for each commercial strain and 2 for BPR), as shown in Table 1. The production cycle was monitored from 17 to 85 wk of age on a 28-d period basis [8].

Collection of Man-Hour Data

To provide producers with a more conclusive comparison of husbandry practices associated with the production environment, the man-hour study consisted of range, cage-free, and cage row replicates as described for experiments 1 and 2. The time spent working within each of the replicates was recorded by personnel to within 1 min. This was converted to man-hours per hen for each of the production periods. The time study records commenced with the first period and consisted of feeding, collection of production records, and all associated hen care procedures were collected each period. The animal caretaker staff used a sign in-sign out record sheet for individual pens, paddocks, or rows; they recorded the actual time going in, completed the required tasks, and then recorded the time out. The number of people required to complete the tasks were documented and time was then multiplied accordingly. Maintenance staff used

the same procedure for the individual replicates to capture all aspects of bird care.

Statistical Analysis

The raw data was summarized by calculating the total time (number of caretakers × minutes) for each replicate. The data were transformed in 2 ways to capture the man-hours per bird. By analyzing the time data based upon the number of hens housed and then by hens surviving, we were able to capture the increase in time due to mortality of the flock. For experiments 1 and 2, all time data was analyzed using PROC GLM in SAS [9]. Means determined to be significantly different were separated using LSMeans/PDIFF.

RESULTS AND DISCUSSION

Experiment 1

Hen Age and Mortality on Man-Hours.

Figure 1 illustrates that care for the range hens housed was higher ($P < 0.0001$) throughout the study. The hens in the range paddocks required more man-hours early in the production phase when the hens were young; man-hours stabilized somewhat by 29 wk of age. This may be indicative of the animal caretakers becoming accustomed to the pen and becoming more efficient in their work patterns. The man-hours from 21 to 61 wk of age were greater ($P < 0.0001$) in the cage-free than in the cage system. This appears to be related to the time where egg production would have been greatest. The time associated with egg collection may have been a contributing factor, as workers had to collect eggs 2 times daily in the cage system during high egg production, whereas once a day the eggs rolled out into egg trays in the nest boxes. Figure 2 illustrates the effect of hen mortality on the man-hours needed in the different production environments. As hens age the man-hours per hen decreases on a hen-housed basis from 0.06 to 0.04 h/bird, but the man-hours per hen surviving increased by 25%. The hours required for hen care were very similar to hen housed shown in Figure 3, except for the range production starting at 53 wk of age, when the man-hours per hen began to reflect the cumulative mortality with significantly increased ($P < 0.01$) man-hours

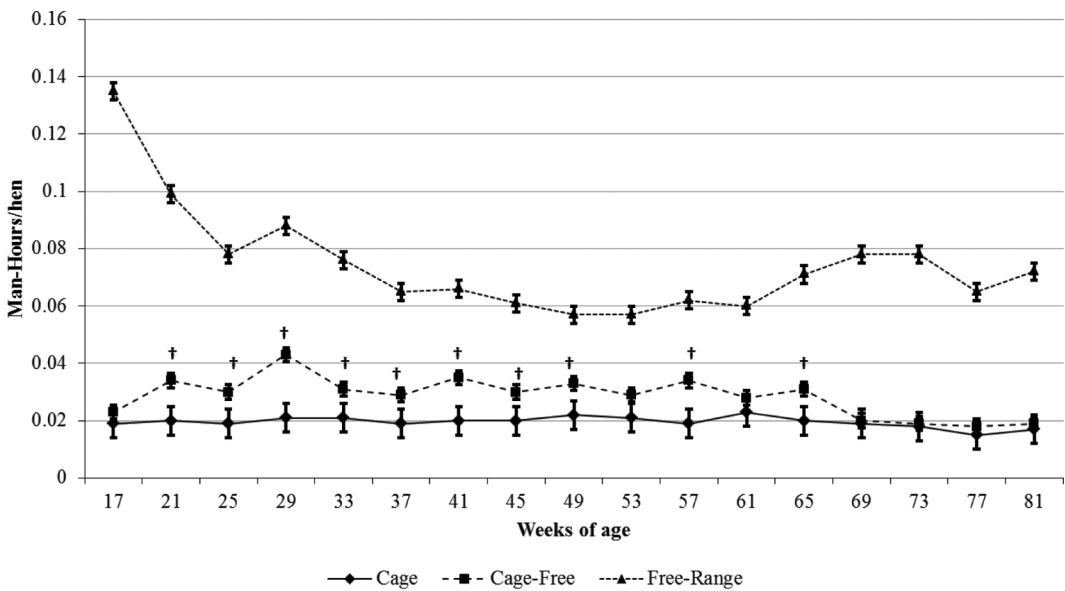


Figure 1. Effect of conventional cage, cage-free, and free-range production environments on man hours per hen housed for hen care as the flock ages in experiment 1. The free-range hens required significantly ($P < 0.0001$) more man hours throughout the study. Symbols († $P < 0.0001$) indicate significant differences between cage and cage-free means.

over the man-hours based on hen housed in subsequent periods through the end of the study. This was due to the cumulative mortality in the later stages of the production cycle. Even though

fewer hens were in the paddocks, the amount of time needed to care for the paddock remained the same, resulting in an increase in man-hours per hen increasing.

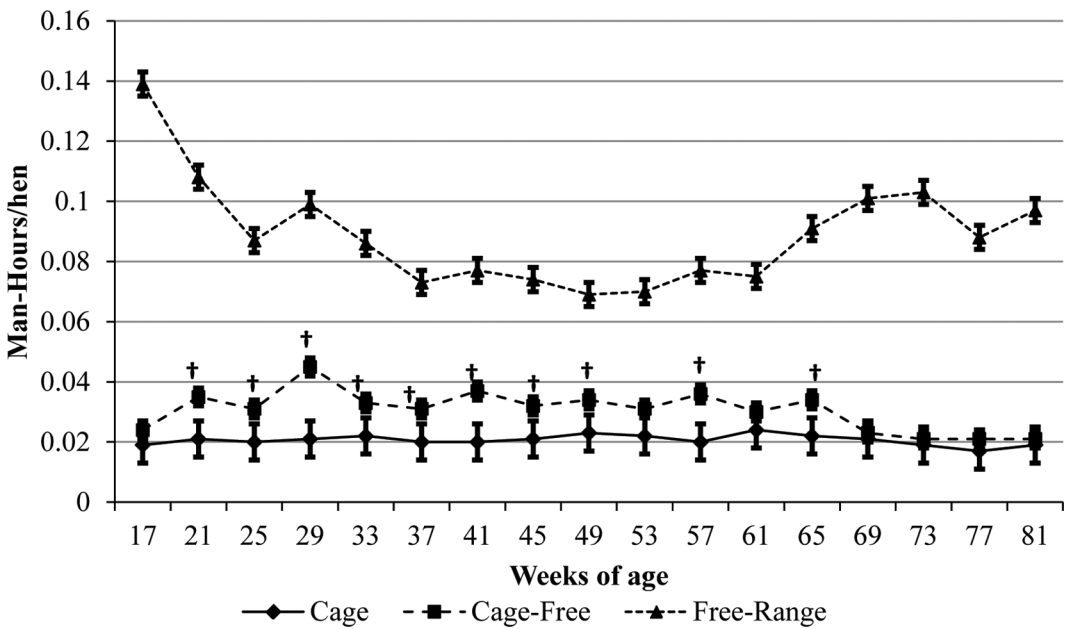


Figure 2. The interaction of mortality and production environment on man hours per hen as the flock ages in experiment 1. The free-range hens required significantly ($P < 0.0001$) more man hours throughout the study. Symbols († $P < 0.0001$) indicate significant differences between cage and cage-free means.

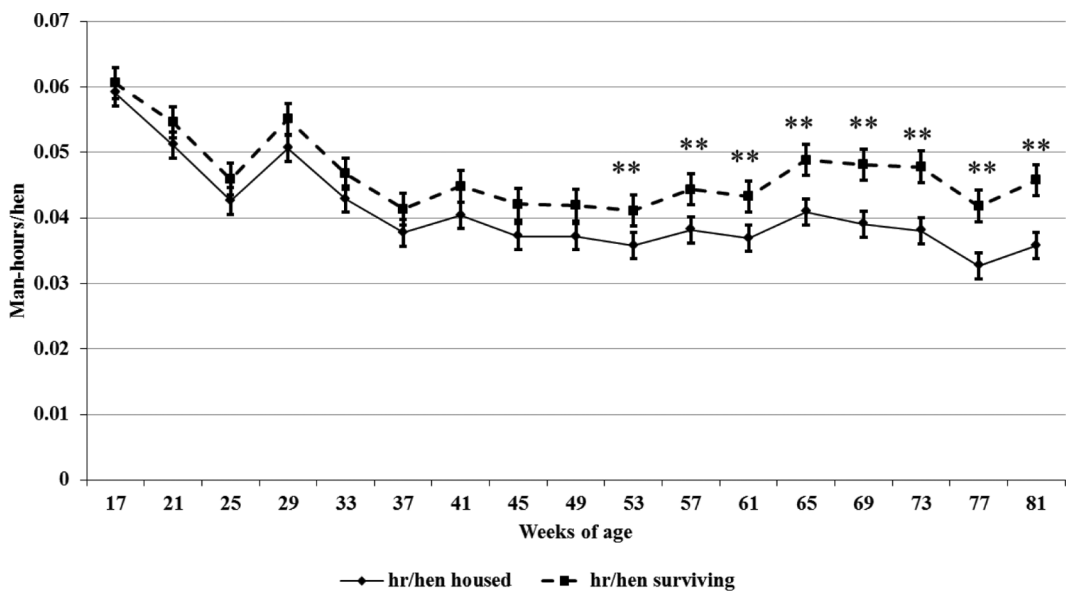


Figure 3. Comparison of hen-housed man hours per hen as compared with surviving hens associated with cumulative mortality in the flock from 17 through 81 wk of age in experiment 1. Symbols (** $P < 0.01$) indicate significant differences between means at the hen ages.

Over the entire production cycle, a significant ($P < 0.0001$) increase of 0.334, 0.486, and 1.268 h/hen was observed in man-hours per hen (Table 2) between the intensive cage system, the less intensive cage-free system, and the extensive range system, respectively. If mortality is taken into account, the rate increases to 0.351, 0.520, and 1.512 h/hen. Initial range research using commercial brown egg strains indicated that man-hours per bird were 0.173 h/bird for cage production and 2.663 h/bird for range production [5]. In this initial time study, the hens were maintained on a static paddock with a single move of the range paddock halfway through the production period; as in the early work [2, 4], the paddock was changed by physically moving the

range hut and the protective fencing. This type of paddock rotation, though effective, was very labor intensive. These man-hours per hen were slightly higher but seem appropriate, as this was a research setting with additional labor inputs. This correlated well with early research from the 1940s and 1950s, which were consistent at approximately 2.6 h per hen for the life of the flock [2, 4]. The reason for the reduction of more than 1 h/hen was the result of the paddock rotation method. In this study we developed a rotation program where we merely had to open and close a gate to switch paddocks every 28-d period. Using this method we were able to maintain good forage cover and allowed for paddock cleansing by rain and sunlight, which helped minimize parasites. This study highlights why complexes gained popularity: the labor input on a per-hen basis continued to decline to about 0.03 h/hen in a 1 million hen complex [10].

Table 2. Effect of the production environment during the laying period on total man-hours per hen in experiment 1

Production environment	Hen housed (h/hen)	Hen surviving (h/hen)
Cage	0.334 ^C	0.351 ^C
Cage free	0.486 ^B	0.520 ^B
Range	1.268 ^A	1.512 ^A
SEM	±0.016	±0.033

^{A-C}Values with different superscripts within columns are significantly different ($P < 0.0001$).

Experiment 2

Effect of Strain on Man-Hours per Hen. The different strains or breeds required similar time commitments when evaluated on a hen-housed basis (Table 3). The HB, HSB, and BPR required similar time commitments of 0.052, 0.051, and 0.049 h/hen per period, respectively.

Table 3. Effect of strain of chicken and mortality on average man-hours per hen within each period

Strain	Hen housed (h/hen)	Hen surviving (h/hen)
Hy-Line Brown	0.052	0.058 ^b
Hy-Line Silver Brown	0.051	0.063 ^a
Barred Plymouth Rock	0.049	0.057 ^b
SEM	±0.001	±0.001

^{a,b}Values with different superscripts within columns are significantly different ($P < 0.001$).

However, when the hour per hen rate is examined for the hens surviving each period, HSB required the greatest ($P < 0.001$) time input over the HB or BPR hens. The HSB required the greatest time associated with hen survival due to the higher mortality rate in the cage-free setting than the other strains [5].

Effect of Management System on Man-Hours per Hen. As in experiment 1, the range hens required the greatest number of man-hours per hen over the course of the production cycle than did the cage-free hens. In both cases, the shift from cage-free to range required approximately 3 times the man-hour input per hen (Table 4). Surprisingly, the difference in egg production [5] between the HB, HSB, and BPR did not result in the man-hour requirement for bird care.

Table 4. Effect of the cage-free and range production environment during the laying period on total man-hours per hen in experiment 2

Production environment	Hen housed (h/hen)	Hen surviving (h/hen)
Cage free	0.459 ^B	0.484 ^B
Range	1.266 ^A	1.530 ^A
SEM	±0.014	±0.033

^{A,B}Values with different superscripts within columns are significantly different ($P < 0.0001$).

An interaction ($P < 0.01$) was observed between the production environment and the strain of hen used on a hen-housed basis (Figure 4). This appears to be related to the response of the HSB hens to the production environments. In the cage-free environment, HB and HSB hens had the highest man-hour needs; however, in the range environment, BPR hens had the highest ($P < 0.01$) man-hour needs. The HB was intermediate in man-hour use; this seems to indicate that the strains used in a particular production system would influence the time needed for hen care.

CONCLUSIONS AND APPLICATIONS

1. Moving from intensive (cage) to extensive production systems requires significant increases in time commitments: a

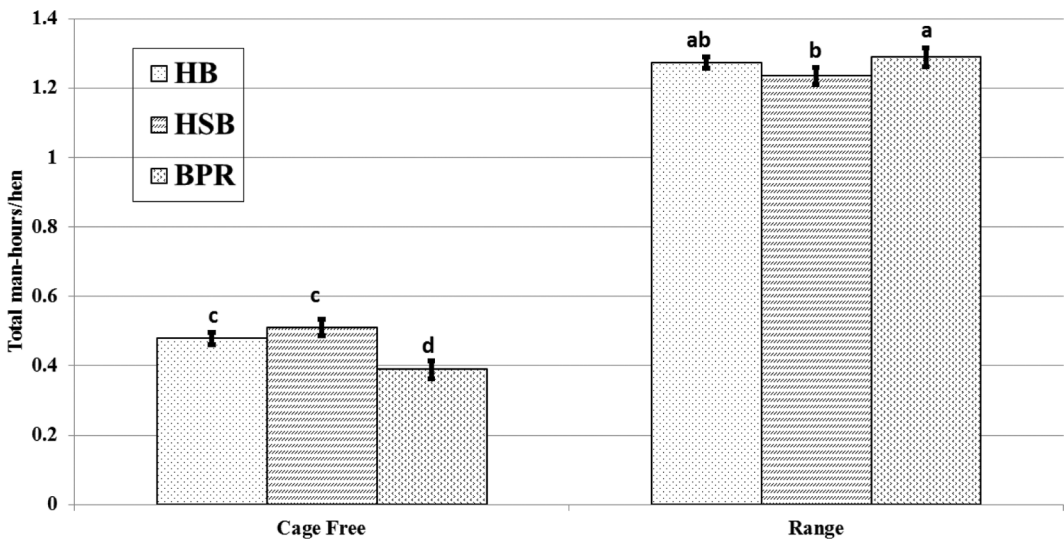


Figure 4. Interaction of production environment and strain on man hours per hen during the entire laying period on a hen-housed basis. Different letters (a–d) indicate significant differences between means ($P < 0.001$). HB = Hy-Line Brown; HSB = Hy-Line Silver Brown; BPR = Barred Plymouth Rock.

45% increase in man-hours from cage to cage free; a 279% increase in man-hours from cage to range; and a 161% increase in man-hours from cage free to range.

2. The man-hours needed for the care of hens in all of the production environments decrease as the hens age.
3. Flock mortality will result in increased man-hours per hen in the later periods of the production cycle
4. The strain of commercial laying hen or the use of a heritage breed may affect the man-hours per hen needed to care for the flock. This may be associated with the productivity of the hens, mortality of the flock, or other production influences of the strain.

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